SOLANA DEVNET / AI-POWERED DEFI

SolanaAI Lend

AI agent autonomously manages a lending protocol in real time.
Smart contract guarantees safety.

Solana Devnet | Gemini 2.0 Flash | 6 ML Models | Anchor/Rust | React | Open Source

Live: http://89.207.255.254

github.com/Islambek201632838/solana-service

Problem: DeFi Lending is Broken

Static Formulas

Aave/Compound use fixed curves that don't react to real-time market conditions

Slow Governance

Parameters change via DAO votes — days or weeks of delay in volatile markets

No Adaptation

Protocols can't respond to sudden crashes, flash loans, or anomalous volume

Result: Cascading liquidations, bad debt, protocol insolvency. Aave lost $1.6M in a single bad-debt event.

DeFi needs real-time intelligence — not governance proposals.

Solution: AI Calculates, Not Guesses

Market Data -> RSI MACD Bollinger -> 6 ML Models -> Sentiment Filter -> Gemini AI -> Validator 11 Rules -> On-chain TX

Gemini = interpreter, not calculator

All numbers come from math formulas and ML models. Gemini only selects from a pre-computed range and explains the reasoning. No hallucinated numbers.

AI acts, not advises

AI sends real Solana transactions — update_parameters, set_sol_price, liquidate, ai_emergency_freeze. Fully autonomous on-chain actions.

Every 2 minutes: CoinGecko price -> 5 indicators -> 6 ML models -> Crash detection -> News sentiment -> Gemini decision -> Validation -> TX to Solana -> Dashboard update

If Gemini is down -> ML-only fallback. If agent dies -> contract auto-rate. Protocol NEVER stops.

AI Intelligence: 6 Models + 3 Engines

ML Models

RandomForest	Trend prediction (up/down/sideways)
IsolationForest	Anomaly detection (flash crashes, pumps)
EWMA Volatility	Regime: low / medium / high / extreme
Crash Detector	6 signals -> probability 0-100%
Risk Scorer	Composite 0-100 from 5 weighted factors
Utilization Predictor	Forecasts pool usage for optimal rate

Smart Features

Dynamic LTV

AI adjusts collateral by volatility
Calm: -5% | Storm: +10% | Extreme: +20%

Credit Score

5 on-chain factors -> 4 tiers
Platinum: -15% collateral, -10% rate

Crash Detection

6 signals -> auto-freeze at >80%
Volume spike, momentum, SMA deviation

Monte Carlo

500 simulations predict liquidation
at 1h / 4h / 24h horizons

Self-Improving

Model reputation tracking
Accurate models gain weight automatically

Sentiment Filter

Noise (Elon tweets) vs Serious (SEC)
Gemini NLP classification

Safety: 7 Layers of Guard Rails

1
Prompt
Gemini gets strict output format + 'ignore noise' instruction

2
Validator
Python checks 11 rules before TX. Confidence < 50 -> rejected

3
Contract
On-chain enforcement: rate [1-20%], collateral [120-200%], cooldown

4
AI Freeze
Auto-freeze when crash probability > 80% or risk score > 90

5
Auto-Rate
If AI agent dies, contract automatically adjusts rates to protect liquidity

6
Insurance
10% of all interest goes to insurance fund to cover bad debt from liquidations

7
Guardrail PDA
Safety parameters stored in separate PDA — only authority can modify. Immutable for AI.

AI can make mistakes. The contract won't allow it.

Every parameter change = on-chain TX, verifiable on Solana Explorer

Why We're Different

Typical AI+DeFi Approach	SolanaAI Lend
Data -> LLM -> "I think maybe..."	Data -> Math -> ML -> Gemini interprets
LLM hallucinates numbers	All numbers from formulas, Gemini only selects
No mathematical foundation	RSI + MACD + Bollinger + ATR + RandomForest
No news filtering	Sentiment: noise -> ignore, serious -> react
AI advises, human acts	AI acts autonomously on-chain (4 TX types)
No guard rails or 1-2 checks	7 layers: prompt -> validator -> contract -> freeze -> auto-rate -> insurance -> PDA
Static model weights	Self-improving: model reputation + dynamic weights
Fixed collateral ratio	Dynamic LTV based on real-time volatility regime

Thank You

AI thinks. Contract controls.

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